

AC-MOT: Adaptive Detector Operating-Point Control for Real-Time Aerial Multi-Object Tracking

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Abstract

Unmanned-aerial-vehicle multi-object tracking must balance observation quality and latency while density, target scale, illumination, and blur vary within a video. We frame AC-MOT as an optimization-driven, training-free detector-control layer that changes confidence, non-maximum-suppression, and input resolution while preserving detector-tracker separation. An early heuristic prototype established the causal interface; V1 was the first formal quality-oriented Optuna calibration, and V2 explicitly optimized the identity-runtime trade-off through a feasible MOTA/IDS Pareto study. Historical transfer and cross-pipeline studies show that the resulting profiles are objective- and host-dependent. We then freeze a modern v3 policy with causal crowd-weighted scene control, recovery, hysteresis, dwell, two tracker hosts, paired bootstrap evidence, and end-to-end timing. On the documented second read of seven VisDrone2019-MOT sequences, our model improves HOTA by 2.937 points with ByteTrack and 1.325 points with OATrack; corresponding MOTA changes are 17.995 and 6.318 points. Tesla T4 throughput is 13.45 and 15.27 frames/s, including approximately 4.3 and 4.2 ms controller overhead. A no-retuning UAVDT transfer is strongly positive for ByteTrack and mixed for OATrack. The evidence supports detector operating-point control and observation-set shaping, but not a separate timing-specific causal gain; these conclusions are bounded by the modern frozen v3 validation protocol.

Keywords: multi-object tracking; UAV video; adaptive inference; real-time image processing

1 Introduction

Multi-object tracking from an unmanned aerial vehicle is an observation problem as much as an association problem. Density, target scale, occlusion, camera motion, illumination, and blur change within a sequence, so a tracker receives an observation stream whose quality and cost are both time-varying [1, 2]. A fixed confidence floor can remove weak but useful boxes in a sparse scene

and retain clutter in a dense one; NMS can merge nearby targets; and fixed resolution spends the same detector budget on easy and difficult frames. These choices affect FP, FN, IDS, HOTA, MOTA, and throughput together.

AC-MOT is a training-free runtime layer that computes inexpensive scene cues, forms a Scene Complexity Index (SCI), and changes detector confidence, NMS, and input resolution while keeping detector and tracker separate. Its scientific

value is the control interface: the detector operating point becomes an auditable runtime variable rather than an immutable deployment setting.

The paper is organized around the optimization-driven evolution of that interface. An early heuristic prototype established the controllable variables and causal boundary. V1 was the first formal quality-oriented optimization; V2 then made identity and deployment cost explicit through a multi-objective Pareto study. Historical transfer and cross-pipeline studies exposed which claims were portable and which were objective- or host-dependent. The modern phase finally froze a stronger detector, a causal policy, a second tracker host, runtime instrumentation, and attribution controls.

The contributions are bounded and evidence-oriented:

1. We distinguish the preliminary heuristic interface from V1, the first formal optimization-derived AC-MOT configuration, and report its exact quality-oriented search and held-out uncertainty.
2. We document V2 as an identity-aware multi-objective extension with an explicit feasible Pareto front and an objective-dependent profile choice.
3. We report historical UAVDT transfer and separate U2MOT/VisDrone and SparseTrack/MOT17 cross-pipeline studies, including their negative or near-neutral evidence.
4. We specify the modern frozen model at implementation level: the crowd-selected SCI, surrounding state logic, thresholds, recovery probe, hysteresis, dwell, causal information flow, and action map.
5. We evaluate ByteTrack and a controlled OATrack host with aggregate metrics and paired 5000-resample sequence bootstrap intervals.
6. We measure end-to-end Tesla T4 latency, no-retuning UAVDT transfer, and matched-static/shuffled-timing attribution controls without claiming a separate timing-specific causal benefit.

The supported claim is precise. AC-MOT is an auditable training-free mechanism for shaping the detector observation stream and computational operating point of aerial MOT. The historical evidence supports distinct quality and

identity/runtime profiles; the frozen modern evidence supports positive changes relative to the declared baseline and strong ByteTrack transfer, with a smaller mixed effect on OATrack. It does not establish universal improvement, official leaderboard equivalence, or that exact scene-conditioned timing is necessary for the observed accuracy gains. Historical, cross-pipeline, and modern protocols are therefore kept separate; extended historical and audit material is provided in Online Resource 1.

2 Related work

2.1 Tracking by detection and aerial MOT

SORT established a compact online pipeline based on motion prediction and assignment, while DeepSORT added appearance information to reduce identity fragmentation [3, 4]. ByteTrack changed the detector-tracker interface by associating high-confidence detections first and then using lower-confidence boxes to recover objects that would otherwise be lost [5]. Other modern trackers further refine motion and appearance association. These trackers primarily improve the interpretation of the boxes they receive; AC-MOT asks the complementary upstream question of which observations, at which resolution and suppression policy, should reach the tracker.

The broader tracking-by-detection literature includes detector-regression tracking (Tracktor) [6], joint detection and association with transformer queries (TrackFormer) [7], and end-to-end alternatives that change the association interface itself. These lines of work motivate treating the detector-tracker boundary as an explicit experimental variable rather than assuming that a tracker can compensate for every detection-side choice. Standardized MOT evaluation and reporting are rooted in MOTChallenge protocols [8], whereas the present aerial studies retain their project-specific protocol boundaries.

VisDrone and UAVDT capture the small targets, changing target count, camera motion, view-point variation, and clutter that make aerial MOT difficult [1, 2]. In this paper they have separated roles: historical VisDrone and UAVDT evidence describes earlier profiles, while a modern frozen protocol tests our model. OATrack is a second

tracker host rather than a claim of superiority over its published system [9]; its frozen repository implementation and tracker-side `min_conf=0.40` are kept distinct from the model’s detector-side confidence floor.

2.2 Detector operating points and adaptive inference

Confidence floors, NMS overlap thresholds, and input resolution are often treated as implementation settings. In aerial scenes they are coupled scientific variables: resolution changes small-object evidence and cost, confidence changes recall and clutter, and NMS changes whether nearby targets remain separate. AC-MOT does not replace detector architecture research; it makes these deployment choices explicit and lets them respond to scene evidence without changing detector weights.

Dynamic inference systems such as Chameleon show that conditional computation can adapt inference to content and resource constraints [10]. Resolution-adaptive networks and spatially adaptive video models likewise use content to trade computation against quality [11]. AC-MOT is intentionally lighter: its state uses scalar image statistics and prior tracker boxes, its action set is finite, and its state machine is auditable line by line. It therefore tests an operating-point intervention at the detector-tracker boundary rather than a learned scheduler.

The detector host is also part of the evidence boundary. YOLOX provides a representative modern anchor-free detector used by the U2MOT cross-pipeline study [12], while U2MOT and SparseTrack illustrate that changes in the association backend can alter the response to the same detector-side policy [13, 14]. OATrack is therefore included as a relevant recent UAV tracker host, not as a published system that this paper claims to outperform [9].

2.3 Optimization and evaluation gap

Black-box configuration is commonly implemented with automated hyperparameter optimization; Optuna’s define-by-run framework and TPE sampling are directly relevant to the V1/V2 development record [15]. V2’s explicit trade-off is reported as a feasible Pareto front, using the

standard notion of non-dominance associated with multi-objective optimization [16]. This distinction matters because V1 and V2 were not two arbitrary manual settings: they answer different constrained questions.

MOTA emphasizes detection and identity errors, IDF1 emphasizes identity precision and recall, and HOTA balances detection and association [17]. Their different responses are useful here because selectivity may reduce FP and IDS while increasing FN. Sequence-level paired bootstrap resampling preserves within-sequence dependence while quantifying system differences [18].

The gap is consequently specific: rigorous evaluation of a training-free online controller that changes detector resolution, confidence, and NMS in aerial MOT while separating historical development from a frozen modern protocol, testing two tracker hosts, measuring runtime, transferring without retuning, and checking whether exact scene timing explains the gain. AC-MOT addresses this gap empirically and does not claim to invent adaptive inference or a new tracker.

3 AC-MOT V1: Quality-Oriented Optimization

An early heuristic prototype was used to identify the controllable detector variables and the detector-tracker causal interface. We define V1 as the first formal optimization-derived AC-MOT configuration evaluated under the frozen experimental protocol. The prototype remains fully documented in Online Resource 1, but it is not treated as the primary AC-MOT generation.

3.1 Optimization design

V1 fixed pretrained YOLOv8n and a tuned ByteTrack profile and searched only on the VisDrone2019-MOT validation split. The test split was not accessed during search. At analysis step j , the controller formed five inexpensive cues: crowd count, the fraction of prior boxes below 32×32 pixels, edge density, low brightness, and low Laplacian variance. Each cue was calibrated empirically from fixed-detector validation outputs or validation image statistics, and the SCI was

their weighted sum,

$$\begin{aligned} S_j &= \sum_{k \in \{c, t, e, n, b\}} w_k x_{k,j}, \\ w_k &\geq 0, \quad \sum_k w_k = 1. \end{aligned} \quad (1)$$

The seven-entry moving average was evaluated every ten frames. Confidence and NMS were linearly interpolated between the selected SCI=0 and SCI=1 endpoints, while two sorted SCI thresholds selected one of the three resolution levels.

The canonical Optuna/TPE study jointly searched five normalized cue weights, the confidence endpoints, the NMS endpoints, and the two SCI thresholds [15]. The resolution levels 512, 928, 960, supported confidence values {0.25, 0.30, 0.35, 0.40, 0.45}, supported NMS values from 0.30 through 0.70, and the temporal settings $W = 7$ and stride 10 were provided by preceding validation screens or the frozen temporal design; they were not falsely presented as variables of the final joint search.

The study recorded the multi-objective directions maximize MOTA, minimize IDS, and maximize FPS. The final validation-only feasibility rule required the FPS gate and $\text{IDS} \leq 271$ (the OLD-A3 validation reference), after which the highest MOTA was selected, with lower IDS, HOTA, IDF1, and FPS as tie breaks. Trial24 was frozen from this procedure. Its exact controller values were

$$\begin{aligned} w_c &= 0.1294928, & w_t &= 0.2217477, \\ w_e &= 0.4337134, & w_n &= 0.0535577, \\ w_b &= 0.1614885, \\ (c_{\text{easy}}, c_{\text{hard}}) &= (0.30, 0.40), \\ (n_{\text{easy}}, n_{\text{hard}}) &= (0.35, 0.35), \\ (\tau_{\text{mid}}, \tau_{\text{high}}) &= (0.1353494, 0.2872868). \end{aligned}$$

3.2 Historical held-out profile

On the historical held-out comparison, the baseline obtained MOTA/HOTA/IDF1 19.729/28.430/32.724, IDS 1235, and 36.528 FPS. V1 obtained 26.948/33.835/41.546, IDS 1184, and 38.985 FPS. The archived paired bootstrap supports V1–baseline gains of +7.2195 MOTA, +5.4051 HOTA, and +8.8220 IDF1, with 95%

Table 1 Optimization-derived historical profiles; full ablation, parameters, bootstrap, and transfer rows are in Online Resource 1, Sec. S1.

Profile	HOTA	MOTA	IDS	FPS
Baseline	28.430	19.729	1235	36.528
V1 quality	33.835	26.948	1184	38.985
V2 identity	31.218	23.792	919	46.024

intervals [5.4362, 9.2934], [4.1273, 6.9022], and [6.9005, 11.0537], respectively. The IDS reduction of 51 has interval $[-114, 219]$ and is therefore not a statistically established identity improvement. V1 is a quality-oriented profile: it was not selected as the lowest-IDS profile and it is not claimed to be a universal operating point.

4 AC–MOT V2: Identity-Aware Multi-Objective Optimization

V1 established that validation-derived detector operating points could improve the historical quality profile, but quality was not the only deployment goal. Identity fragmentation and compute can move in a different direction from MOTA. V2 therefore retained the frozen V1 search design while changing the selection question from a constrained quality profile to an explicit identity–runtime trade-off.

The V2 validation study maximized MOTA and minimized IDS subject only to $\text{FPS} \geq 25$; it removed the V1 $\text{IDS} \leq 271$ gate. It used Optuna’s seeded TPE sampler, validation data only, and the same five-cue SCI, supported detector operating points, seven-entry temporal window, and ten-frame analysis stride. The implementation explicitly constructed a feasible Pareto front: a trial dominated another when it had no lower MOTA and no higher IDS, with at least one strict improvement. The archive records highest-MOTA, lowest-IDS, and balanced Pareto candidates. The balanced choice used the documented equal-weight normalized MOTA/IDS score, with HOTA, IDF1, FPS, and trial index as tie-breaks; Pareto terminology is therefore an implementation fact rather than a retrospective label [16].

Trial22 was the official selected V2 configuration. Its validation record was MOTA 19.330, HOTA 31.651, IDF1 34.226, IDS 168,

and 51.406 FPS. The frozen profile used weights $w_c = 0.1646453$, $w_t = 0.1746265$, and $w_e = 0.5076113$; $w_n = 0.1206956$ and $w_b = 0.0324213$. Its confidence endpoints were 0.40/0.40, NMS endpoints 0.60/0.35, and SCI thresholds 0.2927136/0.6661672, with the same 512/928/960 operating levels. The historical post-selection technical testdev rerun gave MOTA 23.792, HOTA 31.218, IDF1 37.870, IDS 919, and 46.024 FPS; these are reported as a secondary post-hoc record, not as the validation values that selected Trial22.

The V1/V2 distinction is consequently objective-dependent: V1 is the quality-oriented profile, while V2 is the identity/runtime profile. Neither profile replaces the other for every tracker or metric, and V1’s IDS must not be described as worse than the baseline (1184 versus 1235 in the archived held-out record). The complete parameter records, all historical trials, and the uncertainty tables are retained in Online Resource 1.

5 Pre-v3 Generalization Studies

Before the modern freeze, the controller family was examined across datasets and association backends. These studies are kept separate from v3 because their detectors, evaluators, development splits, and action definitions are not identical.

5.1 Historical UAVDT transfer

The frozen historical profiles were transferred to UAVDT without retuning. The baseline obtained HOTA/MOTA/IDF1/IDS 24.085/13.841/27.887/558; V1 obtained 28.390/17.399/34.453/321; and V2 obtained 26.930/16.118/32.014/308. This supports portability of frozen operating profiles, not a universal quality ordering and not evidence that the modern v3 policy was trained on UAVDT.

5.2 U2MOT host on VisDrone

The cross-pipeline study used the published U2MOT host with YOLOX-X [12, 13]. Development used seven VisDrone validation sequences

and held-out evaluation used 17 VisDrone test-dev sequences. The V1 SCI weights and transforms were retained, the tracker was not modified, and the action boundaries were recalibrated from the U2MOT validation distribution using unsupervised 33rd/67th percentiles because the V1 thresholds placed all frames in the hard region. No ground-truth metric was used for that recalibration, but the U2MOT action tuples were hand-set and do not have a selection record.

On test-dev, the custom evaluator reported baseline MOTA 53.9, IDF1 69.8, IDS 1239, FP 41385, and FN 63241; the transferred controller reported MOTA 53.7668, IDF1 69.8494, IDS 1152, FP 40155, and FN 64801. The defensible summary is approximately unchanged accuracy, 87 fewer IDS, 1230 fewer FP, and 1560 more FN. No HOTA, confidence interval, or matched-baseline speed claim is attached. This is a VisDrone cross-pipeline study, not a UAVDT result or a universal quality win.

5.3 SparseTrack host on MOT17

The separate SparseTrack study used the published pseudo-depth association host [14] on MOT17 `val_half`, seven FRCNN sequences, and 2652 frames. Its minimal Adaptive Edge V1 transfer changed only NMS from 0.70 to 0.80; confidence, resolution, detector, and tracker settings were unchanged. The threshold came from a leave-one-sequence-out stump study on those development sequences. Static versus adaptive evaluation gave MOTA 76.8881 versus 76.9252, IDF1 81.4943 versus 81.5299, and IDS 136 versus 123. The MOTA change was only +0.0371 percentage points with bootstrap CI $[-0.041, +0.182]$, so the manifest does not establish superiority. This near-neutral result is retained because it limits claims of broad cross-dataset generalization.

6 Modern Frozen AC-MOT v3 (our model)

6.1 Motivation and development boundary

The modern redesign was needed because the historical controller had been evaluated through several intertwined choices: detector, tracker, operating grid, evaluation protocol, and selection objective. A stronger public YOLO11m checkpoint made detector-side sensitivity a central question, while earlier calibration searches showed that an adaptive policy could fail to beat a matched static point. The modern study therefore froze a detector checkpoint, declared a calibration split, used sequence-level cross-validation, retained negative search results, and evaluated two tracker hosts. It also added bootstrap uncertainty, end-to-end timing, frozen UAVDT transfer, and attribution controls.

Development used eight VisDrone2019-MOT calibration sequences comprising 3282 frames. Four folds held out sequence pairs during calibration; the final seven-sequence modern VisDrone evaluation was not used to choose the policy. The detector is `dronefreak/visdrone-yolo11m`, trained on VisDrone-DET at 640 pixels, with checkpoint hash `c6f98247e7c731a567aaf37ca7b808beff588b687623e028b25a2e1d846e96e7`. This is not the OATrack detector and is not a VisDrone-MOT-trained checkpoint.

6.2 Cue audit and modern SCI

The search first audited which scene information was useful under the new detector and protocol. Stage 3A selected the cue subset `['crowd']`. The result must be stated carefully. The weighted modern SCI is crowd-selected; the wider SceneLayer still computes tiny-object, edge, brightness, and blur statistics because those values participate in scene labels, the tiny guard, and the recovery and hysteresis machinery.

At analysis step j , let N_j be the number of boxes reported by the tracker before the current detector decision. The crowd measurement is

$$C_j = \min(N_j/30, 1). \quad (2)$$

Our weighted index is the mean of the most recent up-to-seven crowd measurements,

$$S_j = \frac{1}{|H_j|} \sum_{k \in H_j} C_k, \quad (3)$$

where H_j is the current seven-entry history. Analyses occur every ten frames. The current frame's tracker output is not available when S_j is computed; it is stored after detection and association for future decisions.

6.3 State target and surrounding guards

The raw target state is HIGH when $S_j > 0.5114928923560124$ or the tiny-object ratio exceeds 0.50. It is MEDIUM when $S_j > 0.29747709701135766$ or the scene label is crowded or tiny. Otherwise it is LOW. HIGH is a state label, not a statement that HIGH uses the largest input resolution.

The target is then processed by the state-stabilization logic. The scene label is **night** when brightness is below 80, **blur** when Laplacian variance is below 180, **tiny** when the tiny ratio exceeds 0.50, and **crowded** when crowd or edge conditions exceed their historical scene thresholds. These labels do not add weight to the selected SCI, but they can guard a state transition. The implementation tracks a decaying peak count. After at least five observed boxes, a count below 0.40 of the decaying peak increments the drop age; during up to three analysis steps after such a collapse, the recovery probe requests HIGH. The probe is a bounded response to a sudden loss of observations, not access to future frames or ground truth.

Downward transitions are held by hysteresis. A HIGH state remains HIGH when $S_j > 0.50$ or tiny exceeds 0.40; a MEDIUM state remains MEDIUM when $S_j > 0.25$. Finally, a 30-frame dwell rule prevents a new target from changing the state within 30 frames of the preceding change. The ordering is target, recovery probe, downward holds, dwell, and state update. The resulting machine is causal and auditable, but it is not asserted to be an optimal controller.

```

1: initialize level LOW, empty seven-entry his-
   tory, empty previous boxes, peak count 0
2: for each frame  $t$  do
3:   if  $t = 1$  or  $t \bmod 10 = 1$  then
4:     compute image statistics and
     crowd/tiny values using boxes from frames
      $< t$ 
5:     append  $C_j$  and set  $S_j$  to the history
     mean
6:     set target HIGH, MEDIUM, or LOW
     using  $S_j$ , tiny, and scene label
7:     update decaying peak; if collapse gate
     passes, apply up to three-step HIGH probe
8:     apply downward holds, then reject
     changes inside the 30-frame dwell interval
9:     update persistent level
10:  end if
11:  select frozen action for the persistent level;
  run detector and confidence filter
12:  update the fixed tracker and store its boxes
  for future frames
13: end for

```

Algorithm 1: Frozen controller used by our model

6.4 Frozen action map

The frozen actions are LOW (1536, 0.70, 0.10), MEDIUM (1280, 0.45, 0.40), and HIGH (1088, 0.60, 0.40), where each tuple is resolution, NMS IoU, and detector confidence floor. The apparent non-monotonicity is intentional. State names describe scene difficulty, whereas the action was selected from the calibration search and need not increase resolution with state rank. Reporting the names as resolution ranks would reverse the meaning of the frozen method.

This algorithm makes the interface explicit. AC-MOT controls the detector operating point; ByteTrack or OATrack remains responsible for association. The detector-side confidence floor is not OATrack’s tracker-side `min_conf=0.40`. No detector scores enter the scene layer, no online labels are used, and no parameter is changed after the freeze.

6.5 Causal execution and implementation boundary

At an analysis frame, the current image supplies edge, brightness, and blur statistics, while tracker boxes from frames $< t$ supply crowd, tiny-object,

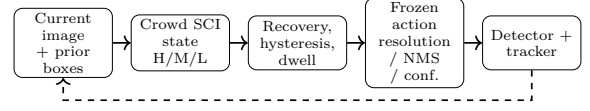


Fig. 1 Frozen state machine for our model; crowd SCI and auxiliary guards remain separate, with causal delayed feedback.

Table 2 Frozen action map for our model. State names are controller labels, not a monotonic resolution ranking.

State	Resolution	NMS IoU	Conf. floor
LOW	1536	0.70	0.10
MEDIUM	1280	0.45	0.40
HIGH	1088	0.60	0.40

and peak-count values. The seven-entry crowd history is updated, the target state is computed, and the collapse probe, hysteresis, and 30-frame dwell are applied in that order. The persistent state then selects the frozen tuple before detector execution; tracker output is stored only for later decisions. Thus the crowd-selected SCI must not be shortened to “crowd-only controller”: tiny guards the high state and recovery, brightness and blur form scene labels, edge can produce a crowded label, and temporal memory remains active. No cue reads detector scores, ground truth, or future frames. The full implementation trace is retained in Online Resource 1, Sec. S3.

LOW, MEDIUM, and HIGH are state names, not an ordered resolution scale. The frozen mapping is LOW 1536, MEDIUM 1280, and HIGH 1088; an implementation must use these tuples rather than substitute a monotonic resolution policy.

7 Experimental protocol

7.1 Historical and modern separation

Historical results come from the legacy freeze: original AC-MOT, V1 Trial 24, V2 Trial 22, historical VisDrone test-dev, and historical UAVDT transfer. Modern results come from tag `acmot-v3-exp-2026-10-04-freeze` and its archived experiment package. The protocols are not pooled because their detectors, operating grids, evaluation details, and held-out roles differ.

7.2 Modern detector, splits, and transfer

The modern detector is `dronefreak/visdrone-yolo11m` trained on VisDrone2019-DET at 640 pixels; the checkpoint SHA256 is `c6f98247e7c731a567aaf37ca7b808beff588b687623e028b25a2e1d846e96e7`. Development uses eight VisDrone2019-MOT calibration sequences and four sequence-level folds. The modern VisDrone read uses seven sequences and 71,830 ground-truth boxes. It is the project’s class-agnostic protocol, not an official VisDrone leaderboard evaluation, and it is disclosed as a second validation read because a static control had already read the sequences.

UAVDT transfer uses 20 sequences, 16,592 frames, and 49,776 detector forward passes from three action-specific cache pairs. The frozen policy is applied without recalibration or retuning, once per tracker host, so no bootstrap or cross-validation claim is attached.

7.3 Trackers, baseline, and metrics

ByteTrack uses the frozen archive defaults. OATrack is the repository’s frozen reimplementation with tracker-side `min_conf=0.40`; this is separate from the model’s detector-side confidence floor. The declared modern baseline is `r1536_n70`: resolution 1536, NMS IoU 0.70, and no detector filter above the cache floor 0.01. Ours uses the frozen action map and thresholds; no detector, tracker, or operating choice changes after the freeze.

We report HOTA, MOTA, IDF1, IDS, FP, FN, precision, and recall. HOTA balances detection and association, MOTA exposes the combined FP/FN/IDS effect, and IDF1 measures identity precision and recall. Baseline and ours are paired by sequence. The archived modern bootstrap uses 5000 paired resamples with seed 42; it is not applied to the single-run UAVDT transfer.

7.4 Runtime and evaluation handling

Our runtime is measured on a Tesla T4 over 200 frames with causal scene analysis, detector execution at the selected resolution, and tracker update. Mean latency, P95 latency, FPS, detector time, tracker time, and controller overhead are

recorded. The baseline harness does not include image-read work identically, so the comparison is reported with that caveat and is not called a perfectly matched speedup. The analysis does not claim a 30-FPS deployment target.

Detector outputs are cached at the required resolution/NMS combinations for reproducibility. The project maps classes to its documented class-agnostic vehicle/tracking protocol before TrackEval-style scoring; these values must therefore not be described as official leaderboard metrics. The archive records the detector hash, controller thresholds, action map, tracker descriptions, calibration folds, forward-pass count, and source checksums. It also records that frozen Trial 7 is not the literal argmax of the documented selection rule under either eligibility interpretation; the paper reports this deviation and does not rerun the search.

8 Results

8.1 Modern VisDrone

Table 3 reports the aggregate results on the documented seven-sequence second validation read. For ByteTrack, our model changes HOTA from 41.586 to 44.523, MOTA from 9.876 to 27.871, IDF1 from 48.569 to 53.929, and IDS from 1341 to 1016. FP falls from 39274 to 24892, while FN rises from 24121 to 25902. This is a selective observation stream: clutter and identity-confusing candidates decrease at a recall cost.

For OATrack, HOTA changes from 46.770 to 48.095, MOTA from 28.165 to 34.483, IDF1 from 57.107 to 59.364, and IDS from 773 to 604. FP falls from 25394 to 21400 and FN changes from 25432 to 25057. The smaller increment is consistent with a stronger baseline and a different association response, not evidence that the controller is ineffective. The two hosts are therefore reported separately rather than pooled into one AC-MOT gain.

The paired bootstrap in Table 4 supports the main positive differences. ByteTrack HOTA, MOTA, IDF1, and IDS changes are +2.937 [2.356,3.375], +17.995 [13.541,23.282], +5.360 [4.490,6.178], and -325 [-498,-164]. OATrack changes are +1.325 [0.200,2.264], +6.318 [4.741,7.879], +2.258 [0.548,3.474], and -169 [-292,-42]. The ByteTrack FN increase has CI

[288,3594.075], whereas OATrack FN has CI [-1462.575,670]; the evidence is consequently a quality trade-off rather than an unqualified recall improvement.

The state occupancy is reported in Online Resource 1, Sec. S2. It confirms that both hosts use multiple actions and that their trajectories differ, but occupancy alone does not establish timing causality. The historical V1/V2 records and the pre-v3 transfer results are reported in their dedicated sections so that their different detectors, splits, and objectives are not pooled with this modern evaluation.

9 Runtime and real-time operating points

The runtime experiment measures scene analysis, state update, detector execution at the selected resolution, confidence filtering, and tracker update on a Tesla T4 over 200 frames. This distinguishes a controller that shapes compute over time from a uniformly larger detector budget.

For ByteTrack, ours has mean latency 74.3238 ms, P95 latency 100.6847 ms, and throughput 13.4546 FPS; detector, tracker, and controller components are 54.5688, 7.9712, and 4.2562 ms. For OATrack, mean latency is 65.4748 ms, P95 85.8969 ms, and throughput 15.2731 FPS; the components are 52.6399, 1.8355, and 4.2010 ms. The detector remains dominant and the controller is small but nonzero, at roughly four milliseconds per frame.

The high-resolution baseline harness reports 139.93 ms and 7.15 FPS for ByteTrack and 128.85 ms and 7.76 FPS for OATrack. Image-read work is not included identically, so this is not a perfectly component-matched speedup. The safe interpretation is reduced measured detector time from the selected actions plus explicit controller overhead. Neither 13.4546 nor 15.2731 FPS is claimed as 30-FPS real-time operation; the P95 values also do not establish a hard per-frame deadline.

The tracker component differs substantially between hosts while detector times are similar, supporting separate tracker-dependent reporting. A future cost-matched study would need equal mean detector work and identical image-read instrumentation to isolate scheduling from average compute.

10 Frozen UAVDT transfer

The modern transfer applies ours to UAVDT without retuning thresholds, weights, action mapping, or tracker configuration. It contains 20 sequences, 16,592 frames, and 49,776 detector forward passes from three action-specific cache pairs. Each host is run once without bootstrap or cross-validation.

For ByteTrack, baseline/ours values are HOTA 45.923/47.004, MOTA -1.573/16.028, IDF1 58.400/61.767, IDS 601/486, FP 279616/212077, and FN 66050/73702. The changes are HOTA +1.081, MOTA +17.601, IDF1 +3.367, IDS -115, and FP -67539; the false-positive reduction accompanies a false-negative increase of 7652. This supports a strong MOTA/precision direction for ByteTrack, not an uncertainty-certified universal improvement.

For OATrack, baseline/ours values are HOTA 47.728/47.640, MOTA 19.607/22.873, IDF1 62.101/62.931, IDS 517/486, and FP 197937/183669. MOTA, IDF1, IDS, and FP improve, but HOTA changes by -0.087. This mixed transfer is why the result is reported per tracker host. The class-agnostic adapter is not an official UAVDT leaderboard protocol, and the VisDrone-trained detector does not make this a detector-domain adaptation result.

11 Discussion and attribution

The modern experiments establish that detector operating point affects downstream MOT, that our model changes the declared high-resolution baseline on two tracker hosts, and that operating-point choices shape measured compute. They do not isolate a separate accuracy contribution from the exact scene-conditioned switching schedule.

The matched-static control used resolution 1088, NMS IoU 0.45, and confidence 0.40. Its HOTA was approximately 44.530 for ByteTrack and 48.269 for OATrack, close to our model's 44.523 and 48.095. The control therefore supports moving away from the declared `r1536_n70` baseline, while leaving the precise timing benefit unresolved. The shuffled-timing control has the same interpretation: its development mean HOTA change was 2.617228 versus 2.446957 for

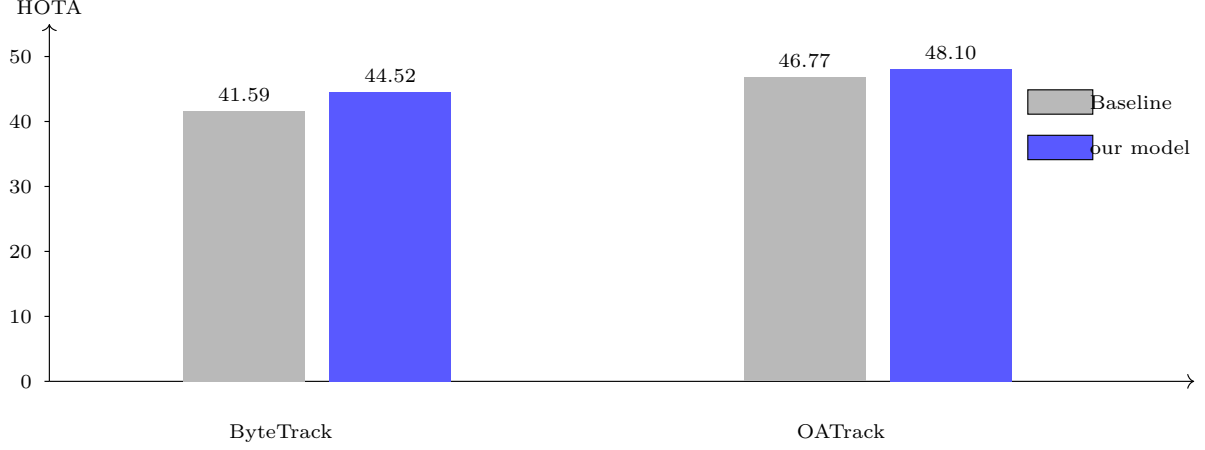


Fig. 2 Modern VisDrone second-read results on seven sequences; hosts are evaluated separately under the custom protocol.

Table 3 Modern VisDrone aggregates on seven sequences and 71,830 ground-truth boxes; our model is a documented second read.

Host	Policy	HOTA	MOTA	IDF1	IDS	FP	FN	Precision	Recall
ByteTrack	baseline	41.586	9.876	48.569	1341	39274	24121	54.849	66.419
ByteTrack	ours	44.523	27.871	53.929	1016	24892	25902	64.852	63.940
OATrack	baseline	46.770	28.165	57.107	773	25394	25432	64.628	64.594
OATrack	ours	48.095	34.483	59.364	604	21400	25057	68.609	65.116

Table 4 Modern paired bootstrap changes versus the predeclared baseline; 5000 paired resamples, seed 42.

Host	Metric	Difference	95% CI
ByteTrack	HOTA	+2.937	[+2.356, +3.375]
ByteTrack	MOTA	+17.995	[+13.541, +23.282]
ByteTrack	IDF1	+5.360	[+4.490, +6.178]
ByteTrack	IDS	-325	[-498, -164]
OATrack	HOTA	+1.325	[+0.200, +2.264]
OATrack	MOTA	+6.318	[+4.741, +7.879]
OATrack	IDF1	+2.258	[+0.548, +3.474]
OATrack	IDS	-169	[-292, -42]

the causal schedule. Detailed control construction and values are retained in Online Resource 1 rather than treated as headline results.

The systems interpretation is still meaningful. A finite policy can allocate detector compute online, react to changing scene statistics, and expose a reproducible operating-point mixture without retraining detector weights. This may be useful under an average detector-time budget even when a matched static point has similar HOTA. It is a deployment interpretation, not evidence that dynamic timing is causally superior.

Tracker dependence reinforces the need to keep hosts separate. ByteTrack starts from a weaker modern baseline and shows the larger VisDrone change; OATrack starts stronger and shows a smaller HOTA increment and essentially flat UAVDT HOTA transfer. The detector-side policy changes the observations available to each association algorithm, so a pooled “AC–MOT gain” would be misleading. The negative searches and the documented Trial 7 selection deviation further bound the conclusion: this paper reports the frozen record and does not rerun the search.

The strongest supported conclusion is that AC–MOT is an auditable, training-free mechanism for shaping the detector observation stream and computational operating point of aerial MOT. It can improve a declared baseline and transfer with tracker-dependent outcomes, but the current evidence does not establish a separate, consistent timing-specific accuracy benefit.

12 Limitations

The modern seven-sequence VisDrone result is a documented second validation read: a static control had already read those sequences before our model. It is therefore a frozen comparison, not an untouched held-out confirmation; the historical test-dev experiment is the cleaner historical held-out evidence. The freeze also records that Trial 7 is not the literal argmax of the documented modern selection rule under either archived eligibility interpretation. We report this provenance limitation without rerunning the search.

Matched-static and shuffled-timing controls weaken a claim that exact switching timing causes the accuracy gain. The project class-agnostic evaluator is not an official VisDrone leaderboard protocol. Only one modern detector checkpoint is used: a third-party VisDrone-DET-trained YOLO11m model rather than a VisDrone-MOT-trained or published OATrack detector. Two tracker hosts show interaction but not tracker independence, and the OATrack implementation is a controlled repository reimplement rather than a direct published-system comparison.

The historical V1/V2 studies use older detectors, custom held-out protocols, and objective-specific profiles. The U2MOT transfer uses a separate VisDrone pipeline with unsupervised boundary recalibration and no HOTA or confidence interval; the SparseTrack transfer is a same-development-set, near-neutral MOT17 study whose bootstrap interval includes zero. Neither establishes universal cross-pipeline quality. UAVDT transfer is a single frozen-policy run per host without cross-validation or bootstrap, and OATrack HOTA is essentially flat there.

T4 timing is measured, but image-read work is not instrumented identically in the baseline and our harnesses. Finally, throughput is below 30 FPS. These limits constrain generalization and attribution while leaving a reproducible detector-operating-point study with measured quality, cost, and transfer behaviour.

13 Conclusion

AC-MOT is best understood as an optimization-driven detector operating-point interface rather than as a single timeless heuristic. An early prototype established the causal boundary. V1

was the first formal quality-oriented optimization and produced a quality profile with uncertainty. V2 then made identity and deployment cost explicit through a feasible Pareto search, showing why a lower-IDS/faster profile need not be the strongest HOTA profile. Historical UAVDT transfer, U2MOT on VisDrone, and SparseTrack on MOT17 further showed that profile choice and tracker host matter, including negative and near-neutral cases.

The modern frozen AC-MOT v3 phase tightened the protocol around a stronger detector, a crowd-selected SCI with causal guards, a finite action map, two tracker hosts, paired bootstrap evidence, T4 instrumentation, and no-retuning UAVDT transfer. The modern results support positive changes over the declared `r1536_n70` baseline on ByteTrack and OATrack, with stronger ByteTrack effects and mixed OATrack transfer.

The final conclusion remains bounded. Matched-static results are close to our model and shuffled timing is at least as good on calibration, so the current record does not establish that exact scene-conditioned timing is necessary for the observed accuracy gains. It supports training-free detector operating-point control as a reproducible mechanism for shaping tracking quality and compute, while timing-specific causality, universal tracker independence, and broader detector generalization remain questions for explicitly designed future experiments.

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Ethics approval and consent to participate. This computational study uses public benchmark data and does not involve human participants or animals.

Consent for publication. Not applicable.

Data and code availability. The frozen controller specification, result records, evidence map, staged sections, scripts, and build package are included in the project repository. Dataset redistribution remains subject to the original benchmark licenses and access terms.

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